

Peter Sauer

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Portfolio: petercsauer.github.io • GitHub: github.com/petercsauer • LinkedIn: linkedin.com/in/petercsauer

5+ years across the full stack. Embedded firmware, mechanical design, distributed systems, perception. Just as comfortable soldering in the field as architecting comms systems in Docker. Shipped 50+ autonomous vessels internationally for defense and maritime customers.

EDUCATION

Master of Science in Robotics Northwestern University,
2020 - 2021

BS Mechanical Engineering, Minor in CS University of
Wisconsin, 2016 - 2020

WORK EXPERIENCE

Senior Software Engineer — Applied Intuition (Remote) Aug 2024 - Present

- Built a dockerized multi-protocol comms system (gRPC, ZeroMQ, MQTT, UDP, ROS2) for heterogeneous swarm control. Tested with 50+ agents
- Ran full autonomy stack with perception inference on 50 physical vessels on water
- Took perception pipeline from 5 FPS to 30 FPS. Cut container size from 20 GB to ~5 GB
- Built real-time sensor processing pipelines with ONNX Runtime and Triton Inference Server
- Built internal GUIs for swarm command and control. Became the team's primary operator interface

Senior Mechatronics Engineer — Ocean Power Technologies Aug 2023 - Aug 2024

- Owned architecture decisions (software, firmware, hardware) for 8-ft, 16-ft, and 22-ft WAM-V platforms. Team of 4
- Helped cut vessel build time from 6 months to 2 months with better documentation and integration processes
- Mentored 4 junior engineers on field deployment and system integration
- Deployed to 6+ locations: Bahrain (mine countermeasures), Alaska/Texas/Florida (bathymetric survey), San Diego/Port Hueneme (surveillance)

Mechatronics Engineer — Ocean Power Technologies Sep 2021 - Aug 2023

- 5th employee. Grew through an acquisition to 20+ people, shipped 35-40 vessels total
- Designed embedded firmware controlling propulsion, steering, vision, and comms (GPS, Iridium, Starlink, cellular, mesh radios) across several embedded platforms, enabling fully autonomous vessel operation
- Built onboard software for autonomous navigation, surveillance, and seafloor mapping
- Led field integration for 10+ vessels across 6 deployment sites, including first autonomous mine countermeasures demo in Bahrain and Arctic bathymetric surveys

Mechanical Engineer — Presco Inc. Mar 2020 - Aug 2020

- Prototyped a UV-C mask cleaner with Arduino-based safety timer (COVID-era project)
- Built a UV-C mail sanitizer with dual stepper motors for automated feed-through

Creative Lead — Project Cyclone 2020 - 2021

- Co-founded predictive maintenance startup. Designed the UI for an ML platform that generates maintenance schedules from machine sensor data
- Won multiple startup competitions, secured \$50K in funding, and shipped a working MVP of the predictive maintenance platform

SKILLS

Programming: Python, C/C++, JavaScript, TypeScript, HTML, CSS, React, Git, CI/CD, Linux Systems Administration

Robotics & Infrastructure: ROS/ROS2, Docker, gRPC, ZeroMQ, MQTT, UDP, Protocol Buffers, Distributed Systems

ML & Perception: ONNX Runtime, Triton Inference Server, Computer Vision Pipelines, Sensor Fusion

CAD & Design: SolidWorks, Autodesk Inventor, Onshape, Adobe Illustrator, Photoshop, 3D Printing

EXTRACURRICULARS

Wisconsin Sailing Team Captain —

Managed 40-50 sailors competing nationally. \$100K+ budget, travel coordination, hiring.

FIRST Tech Challenge Founder —

Founded FTC team 6081; World Championship awards three years running.